MD. NAJMUL HOSSAIN NUR

CSE Undergraduate, United International University, Dhaka, Bangladesh

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Research Interests

Machine Learning - Computer Vision - Autonomous Systems & UAVs - Embedded Systems & Robotics - Full-Stack Web Development - UI/UX Design

Education

United International University, Dhaka, Bangladesh *2022 — Expected July 2027*

Bachelor of Science in Computer Science and Engineering — Current CGPA: 2.87/4.00

Collectorate School & College, Rangpur *2019 — 2021*

Higher Secondary Certificate (HSC), Science

Lions School and College, Rangpur *2017 — 2019*

Secondary School Certificate (SSC), Science

Research Experience

AgriML-BD — Crop Yield Prediction and Recommendation Using Machine Learning for Bangladesh Agriculture

- Supervised by: Ms. Sadia Islam, Assistant Professor, Dept. of CSE, UIU.
- Ongoing
- We present an ML system for Bangladeshi farmers combining crop recommendation and yield prediction, deployed as a Flask web application, comparing Random Forest, Gradient Boosting, and Deep Neural Network models across 18 experiments.

Robotics & UAV Team Experience

Uro Bangladesh — Student-led Robotics & UAV Research Team, UIU *Member*

- Member of UIU's student-led robotics and UAV research team that competed in the Rotary Wing category at the **TEKNOFEST 2026 International UAV Competition** with **Prohori**, an autonomous rotary-wing UAV built to carry out complex aerial missions with minimal human intervention.
- Qualified as an international finalist through the Project Presentation Report and Mission Video stages, then completed technical checks, UAV preparation, field testing, and mission flights in Malatya, Türkiye (September 8–13, 2026), among 113 competing teams.
- Placed **10th globally and 2nd in Asia**.

Projects

Smart Waste Management System (GitHub) *Electronics Lab · Team Leader*

Bluetooth-controlled robotic waste collection system with ultrasonic detection, robotic arm/gripper control, and real-time LCD monitoring via a MIT App Inventor app.

Tech Stack: Arduino Mega 2560, HC-05, HC-SR04, Servo Motors, 16x2 LCD.

Autonomous Underwater Monitoring Submarine (AUMS) (GitHub) *Micro Lab · Team Leader*

Underwater monitoring robot with navigation control and sensor integration; automatically searches for a target and sends captured images to the website.

Tech Stack: Raspberry Pi 5, ESP32, sensor & motor integration.

Sohaj Jogan — Micro E-commerce Platform (GitHub) *Web Project · Team Leader, Light Seekers*

Location-based smart shopping platform connecting buyers with nearby local shops, with live inventory, delivery tracking, and in-app messaging.

Tech Stack: HTML, CSS, JavaScript, PHP, MySQL, Leaflet.js.

UNO Night (JavaFX Multiplayer Game) (GitHub) *AOOP Lab · Team Leader, Light Seekers*

Realtime multiplayer UNO game with four game modes, in-game currency and shop, clan system, and Wi-Fi/LAN-based socket networking.

Tech Stack: Java, JavaFX GUI, Socket Programming.

SearchAR — Community Safety Platform (GitHub) *SE Lab · Team Leader, Light Seekers*

AI-powered platform for missing-person detection, fire hazard detection, and crowdsourced crime reporting using deep learning and computer vision, with multi-role dashboards for Public, Volunteers, Police, and Admins.

Tech Stack: Deep Learning, Computer Vision, Figma (UI/UX).

PlantGuard (GitHub) *AI Lab · Team Project*

AI-powered Flask web application that detects plant diseases from images and suggests treatments using deep learning.

Tech Stack: Python, Flask, Deep Learning.

Restaurant Management System (GitHub) *Web Lab · Team Project*

PHP/MySQL platform streamlining restaurant operations end-to-end, from table booking to kitchen order tracking, with role-based dashboards for Admin, Chef, Waiter, and Customer.

Tech Stack: PHP, MySQL.

Design / UI-UX

- Designed and prototyped the SearchAR mobile and web app flows in Figma (emergency rescue UI, SOS navigation, dashboard flows) — Figma Link
- Designed the Restaurant Management System web prototype in Figma, focused on clean dashboards and quick order actions — Figma Link

Technical Skills

Programming: Java, C, C++, Python

Web Development: HTML, CSS, JavaScript, PHP, React, Node.js

Database: MySQL, SQL, PostgreSQL

Frameworks & Tools: JavaFX, Figma, Scene Builder, Git & GitHub

IoT / Embedded / Design: Arduino Mega, ESP32, Raspberry Pi 5, Sensor & Motor Integration, Fusion 360

Languages: Bengali (First Language), English (Advanced, C1)

Achievements

- 10th Globally & 2nd in Asia — TEKNOFEST 2026 International UAV Competition, Rotary Wing category (Malatya, Türkiye), as a member of Uro Bangladesh, UIU
- Champion — UIU CSE Project Show, Spring 25 (Database Management Systems Project)
- 4th Place — UIU CSE Project Show, Fall 24 (Advanced Object-Oriented Programming Project)
- 4th Place — UIU CSE Project Show, Summer 25 (Systems Analysis and Design Project)
- Top 10 — UIU CSE Project Show, Spring 26 (Software Engineering Project) for SearchAR

Extra Curriculars

- Member — Uro Bangladesh, UIU's student-led robotics and UAV research team (TEKNOFEST 2026 finalist, 10th globally)
- Team Leader — Smart Waste Management System; Autonomous Underwater Monitoring Submarine; Light Seekers project team (Sohaj Jogan, UNO Night, SearchAR)
- Videographer & Photographer — Filmed and edited project videos and photos; media published on Facebook: UniqueClickerBD